CS 412 / 512
Introduction to Computer Graphics

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COMPUTER SCIENCE

Mathematical Foundations for Computer Graphics

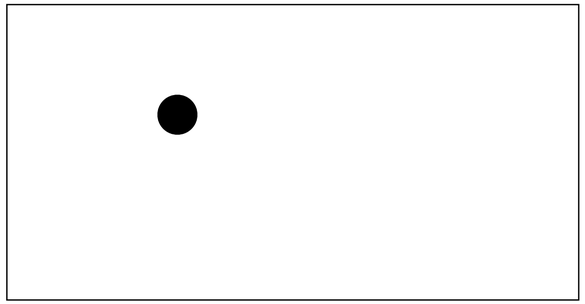
Point

A location in a coordination system

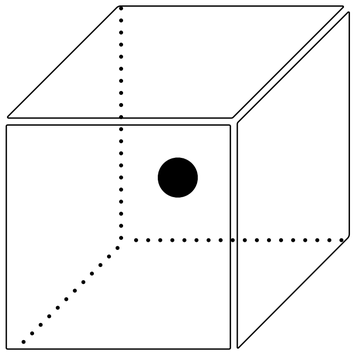
1D P = (x)



2D P = (x, y)



3D P = (x, y, z)



Vectors

Direction and magnitude

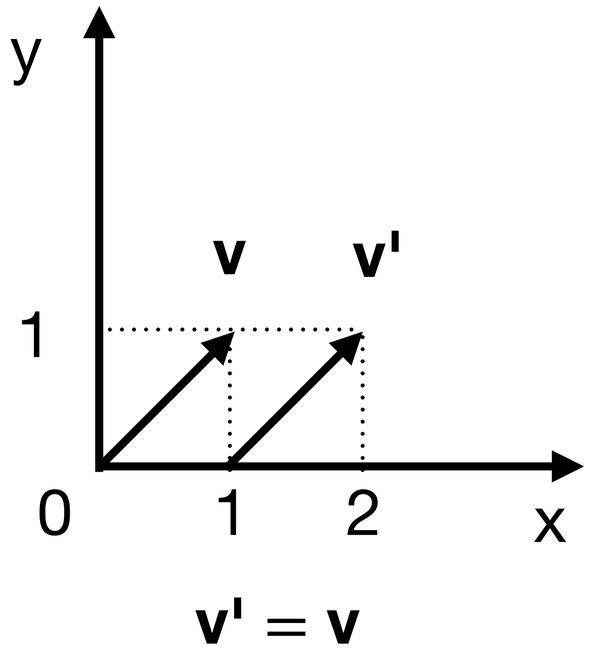
1D [x]

2D [x, y]

3D [x, y, z]

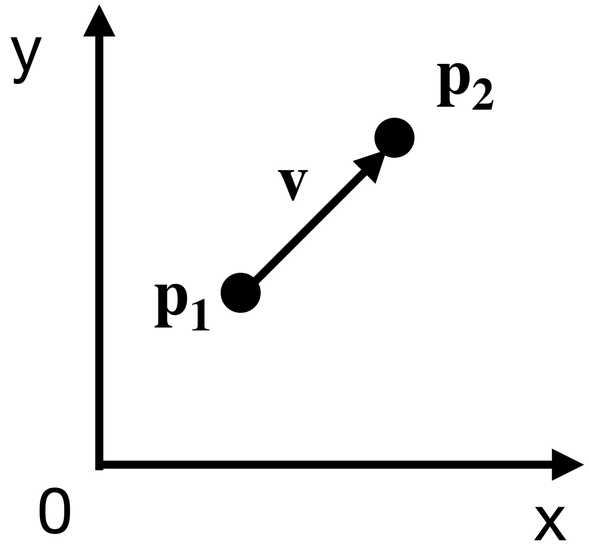
4D [x, y, z, w]

nD $[v_1, v_2, v_3, \ldots, v_n]$



Vectors

$$\overrightarrow{v} = \mathbf{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = [v_x \; v_y \; v_z]^T$$



$$\mathbf{v} = \mathbf{p_2} - \mathbf{p_1}$$

Vectors

Length and normalization

Vector length

$$\|\mathbf{v}\| = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

Unit vector

$$\hat{v} = \frac{\mathbf{v}}{\|\mathbf{v}\|} \qquad \|\hat{v}\| = 1$$

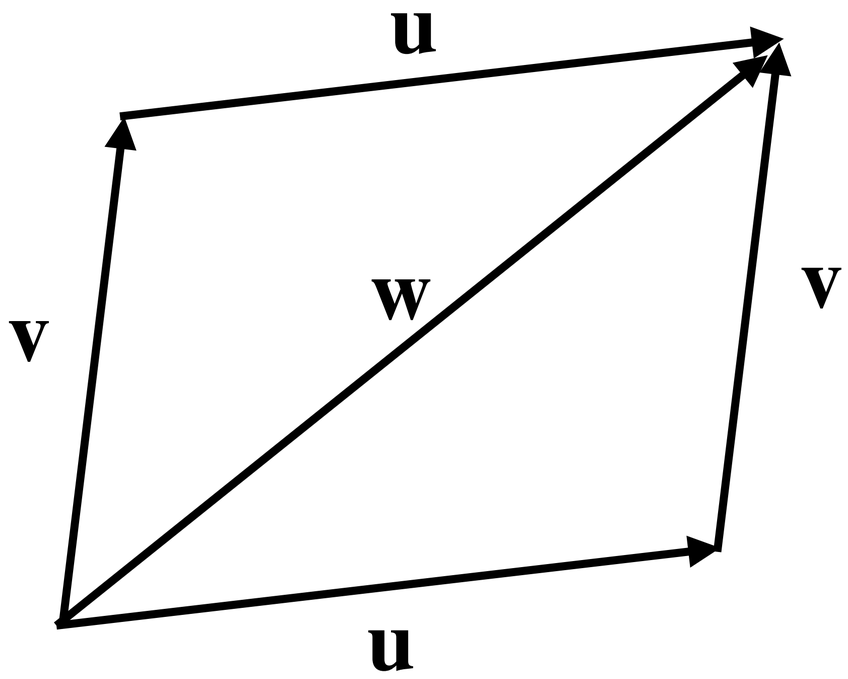
Normalized to a unit vector with length 1 and the same direction.

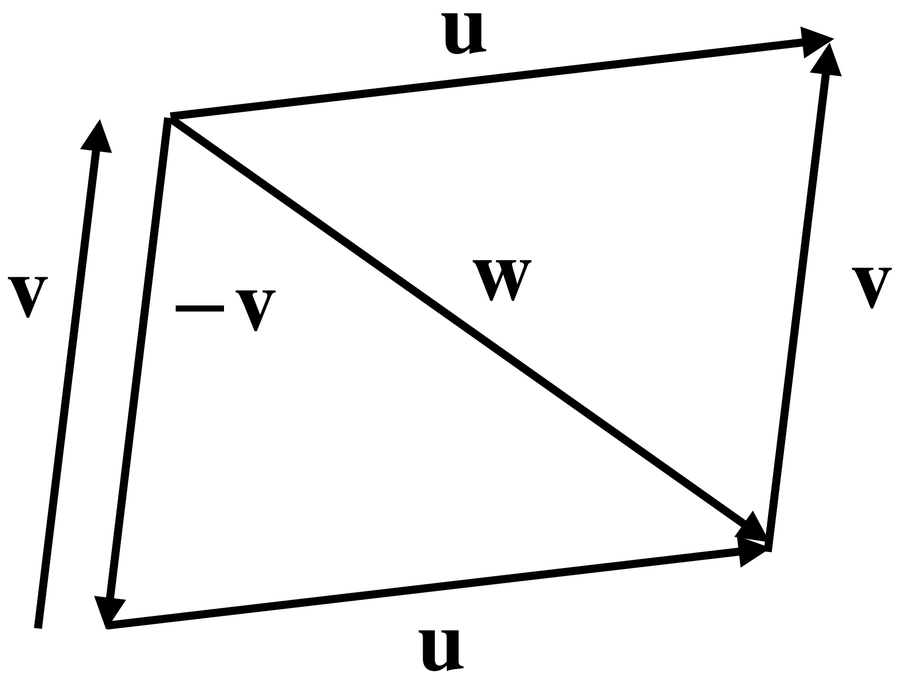
Vectors

Addition and subtraction

$$\mathbf{w} = \mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u} = \begin{bmatrix} u_x + v_x \\ u_y + v_y \\ u_z + v_z \end{bmatrix}$$

$$\mathbf{w} = \mathbf{u} - \mathbf{v} = -\mathbf{v} + \mathbf{u} = \begin{bmatrix} u_x - v_x \\ u_y - v_y \\ u_z - v_z \end{bmatrix}$$

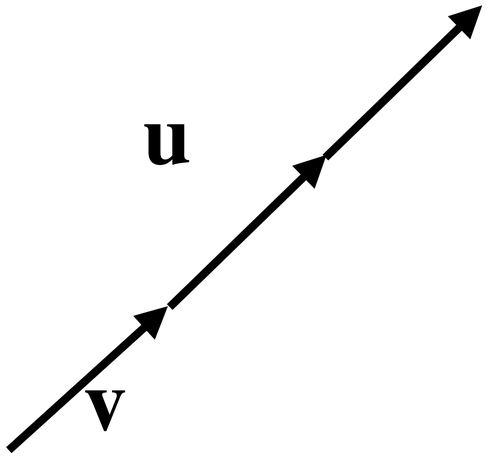




Vectors

Scalar multiplication

$$\mathbf{u} = c\mathbf{u} = \begin{bmatrix} cv_x \\ cv_y \\ cv_z \end{bmatrix}$$



$$\mathbf{u} = 3\mathbf{v}$$

|c| > 1, stretched

0 < |c| < 1, shrunk

c = 1, unchanged

c = 0, zero vector

c < 0, direction flips

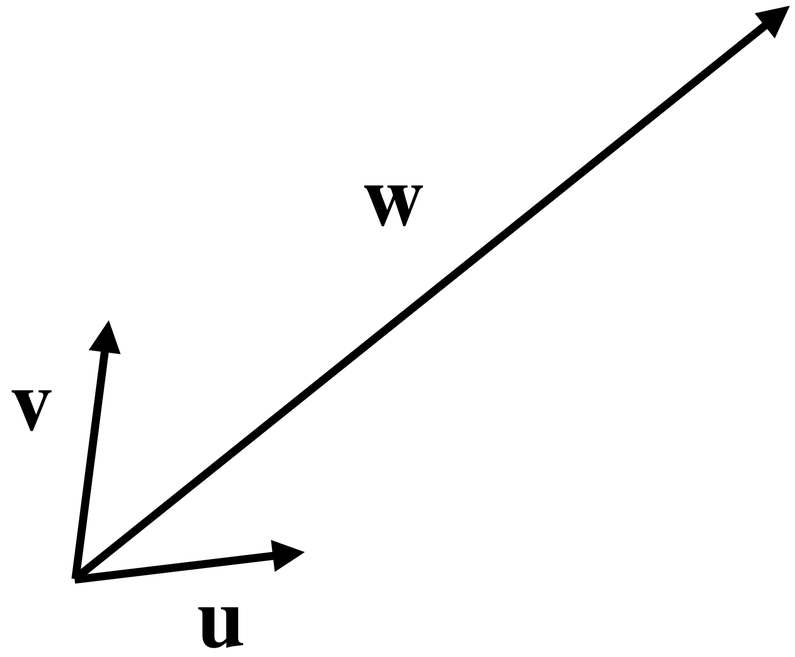
The scalar scales length, but the direction keeps the same or flipped

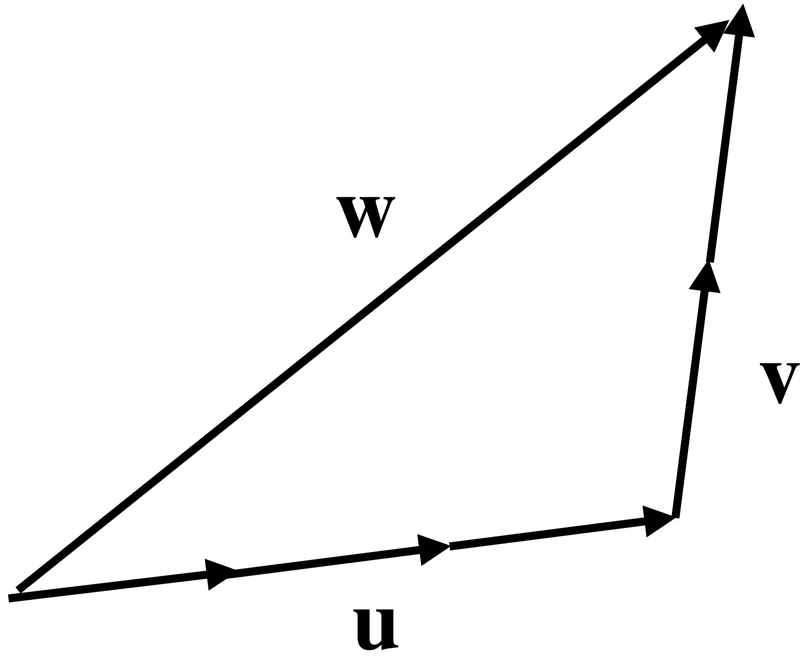
Vectors

Linear combination

$$\mathbf{w} = a\mathbf{u} + b\mathbf{v} = \begin{bmatrix} au_x + bv_x \\ au_y + bv_y \\ au_z + bv_z \end{bmatrix}$$

$$\mathbf{w} = 3\mathbf{u} + 2\mathbf{v}$$



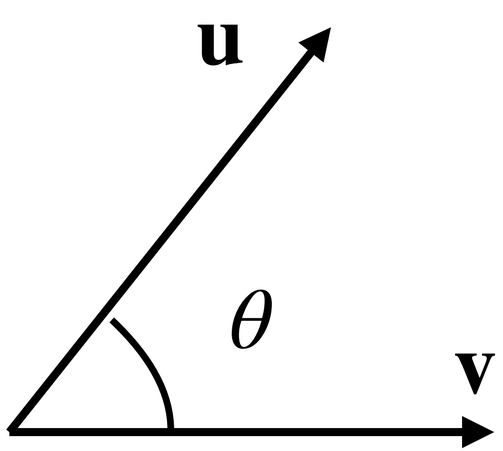


Dot product

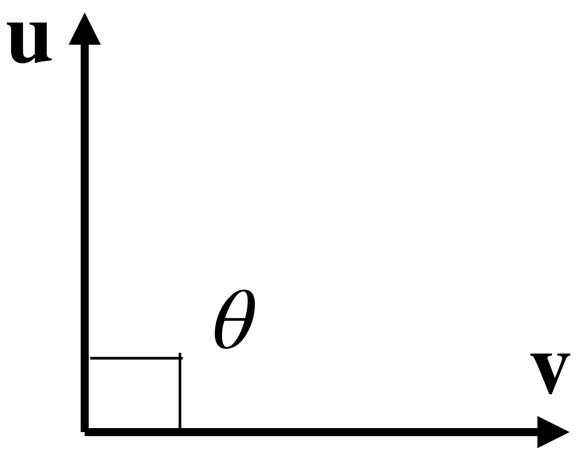
$$\mathbf{u} \cdot \mathbf{v} = u_x v_x + u_y v_y + u_z v_z$$

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| cos\theta$$

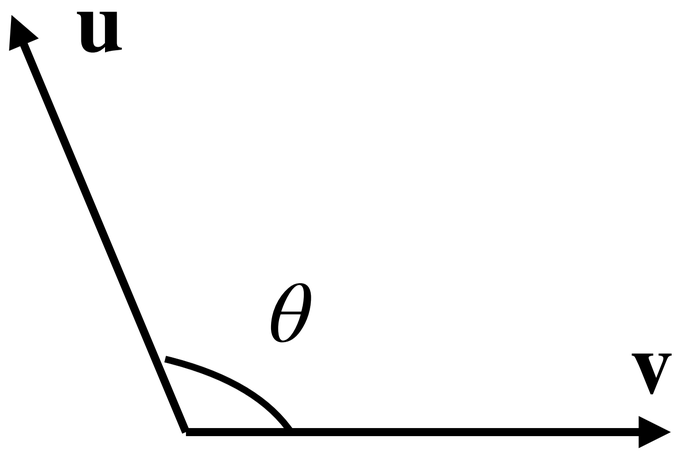
$$cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$



$\mathbf{u} \cdot \mathbf{v} > 0$

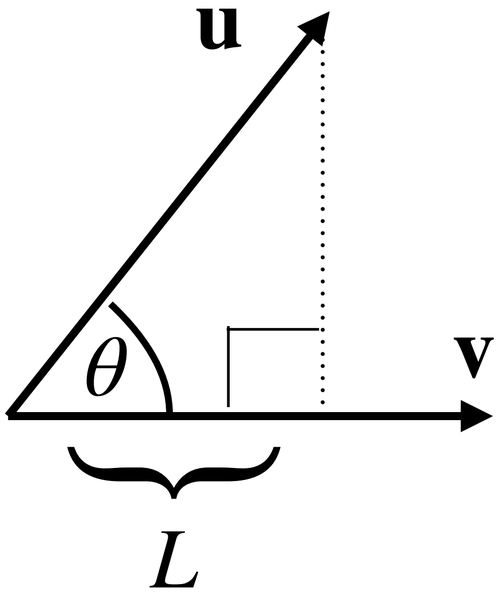


$\mathbf{u} \cdot \mathbf{v} = 0$



$\mathbf{u} \cdot \mathbf{v} < 0$

Length of projection



$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| cos\theta$$

$$L = \|\mathbf{u}\| cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}$$

Algebraic Properties

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$$

$$\mathbf{w} \cdot (\mathbf{u} + \mathbf{v}) = \mathbf{w} \cdot \mathbf{u} + \mathbf{w} \cdot \mathbf{v}$$

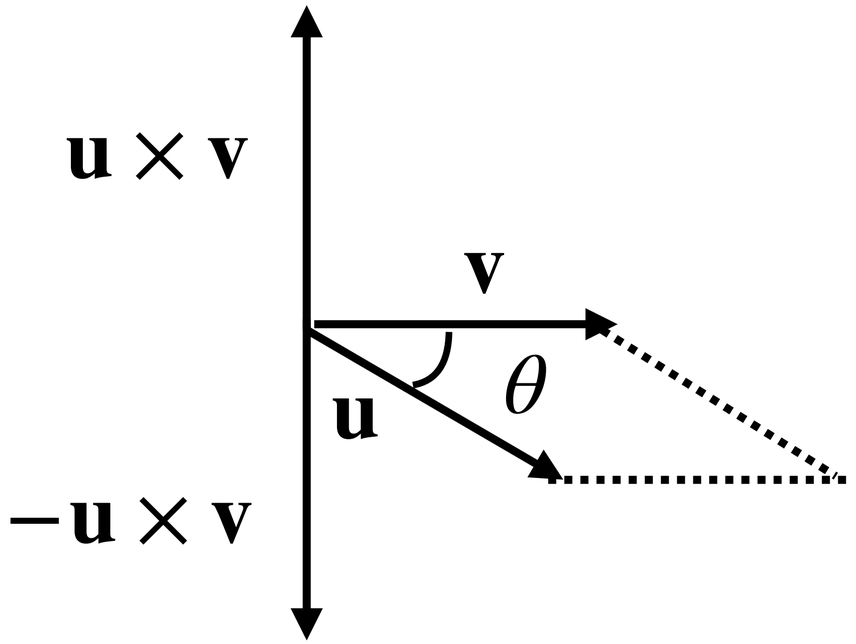
$$\lambda\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \lambda\mathbf{v} = \lambda(\mathbf{u} \cdot \mathbf{v})$$

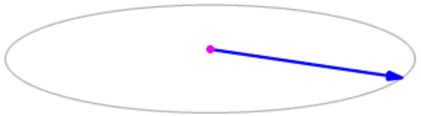
Cross product

$$\mathbf{u} \times \mathbf{v} = det \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix} = (u_y v_z - u_z v_y, u_z v_x - u_x v_z, u_x v_y - u_y v_x)$$

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| sin\theta$$

$$sin\theta = \frac{\|\mathbf{u} \times \mathbf{v}\|}{\|\mathbf{u}\| \|\mathbf{v}\|}$$





[Source:Wikipedia]

Algebraic Properties

$$\mathbf{u} \times \mathbf{v} = -\mathbf{v} \times \mathbf{u}$$

$$\mathbf{w} \times (\mathbf{u} + \mathbf{v}) = \mathbf{w} \times \mathbf{u} + \mathbf{w} \times \mathbf{v}$$

$$\lambda\mathbf{u} \times \mathbf{v} = \mathbf{u} \times \lambda\mathbf{v} = \lambda(\mathbf{u} \times \mathbf{v})$$

Matrices

2D → 2 x 2

3D → 3 x 3

4D → 4 x 4

nD → n x n ⟶ $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$

Matrices

Addition and subtraction

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{11} \\ a_{21} & a_{22} & a_{22} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad \mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \qquad \mathbf{A} - \mathbf{B} = \begin{bmatrix} a_{11} - b_{11} & a_{12} - b_{12} & a_{13} - b_{13} \\ a_{21} - b_{21} & a_{22} - b_{22} & a_{23} - b_{23} \\ a_{31} - b_{31} & a_{32} - b_{32} & a_{33} - b_{33} \end{bmatrix}$$

Matrices

Scalar Multiplication

$$k\mathbf{A} = \begin{bmatrix} ka_{11} & ka_{12} & ka_{11} \\ ka_{21} & ka_{22} & ka_{22} \\ ka_{31} & ka_{32} & ka_{33} \end{bmatrix}$$

Matrices

Vector-Matrix Multiplication

$$\mathbf{A} \cdot \mathbf{v} = \begin{bmatrix} a_{11} & a_{12} & a_{11} \\ a_{21} & a_{22} & a_{22} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \cdot \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = \begin{bmatrix} a_{11}v_x + a_{12}v_y + a_{13}v_z \\ a_{21}v_x + a_{22}v_y + a_{23}v_z \\ a_{31}v_x + a_{32}v_y + a_{33}v_z \end{bmatrix}$$

Matrices

Vector-Matrix Multiplication

$$\mathbf{A} \cdot \mathbf{v} = \begin{bmatrix} a & a_{12} & a_{11} \\ a_{21} & a_{22} & a_{22} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \cdot \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = \begin{bmatrix} a_{11}v_x + a_{12}v_y + a_{13}v_z \\ a_{21}v_x + a_{22}v_y + a_{23}v_z \\ a_{31}v_x + a_{32}v_y + a_{33}v_z \end{bmatrix}$$

$$\begin{bmatrix} | \\ c \\ | \end{bmatrix} = \begin{bmatrix} -a_1- \\ -a_2- \\ -a_3- \end{bmatrix} \begin{bmatrix} | \\ b \\ | \end{bmatrix}$$

$$c_i = a_i \cdot b$$

$$\begin{bmatrix} | \\ c \\ | \end{bmatrix} = \begin{bmatrix} | & | & | \\ a'_1 & a'_2 & a'_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\mathbf{C} = a'_1 \cdot b_1 + a'_2 \cdot b_2 + a'_3 \cdot b_3$$

Matrices

Matrix-Matrix Multiplication

$$C = AB = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

$$\mathbf{A} \cdot \mathbf{B} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} & a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} & a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{33} \end{bmatrix}$$

Matrices

Properties of matrix multiplication

$$\mathbf{AB} \neq \mathbf{BA}$$

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$$

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC} \qquad (\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$$

Matrices

Matrix-Matrix Multiplication

Can we multiply any two arbitrary matrices?

The number of columns in the first matrix has to be the same with the number of rows in the second!

$$\mathbf{A} \in \mathbb{R}^{m \times n} \qquad \mathbf{B} \in \mathbb{R}^{n \times p} \qquad \mathbf{AB} \in \mathbb{R}^{m \times p}$$

Matrices

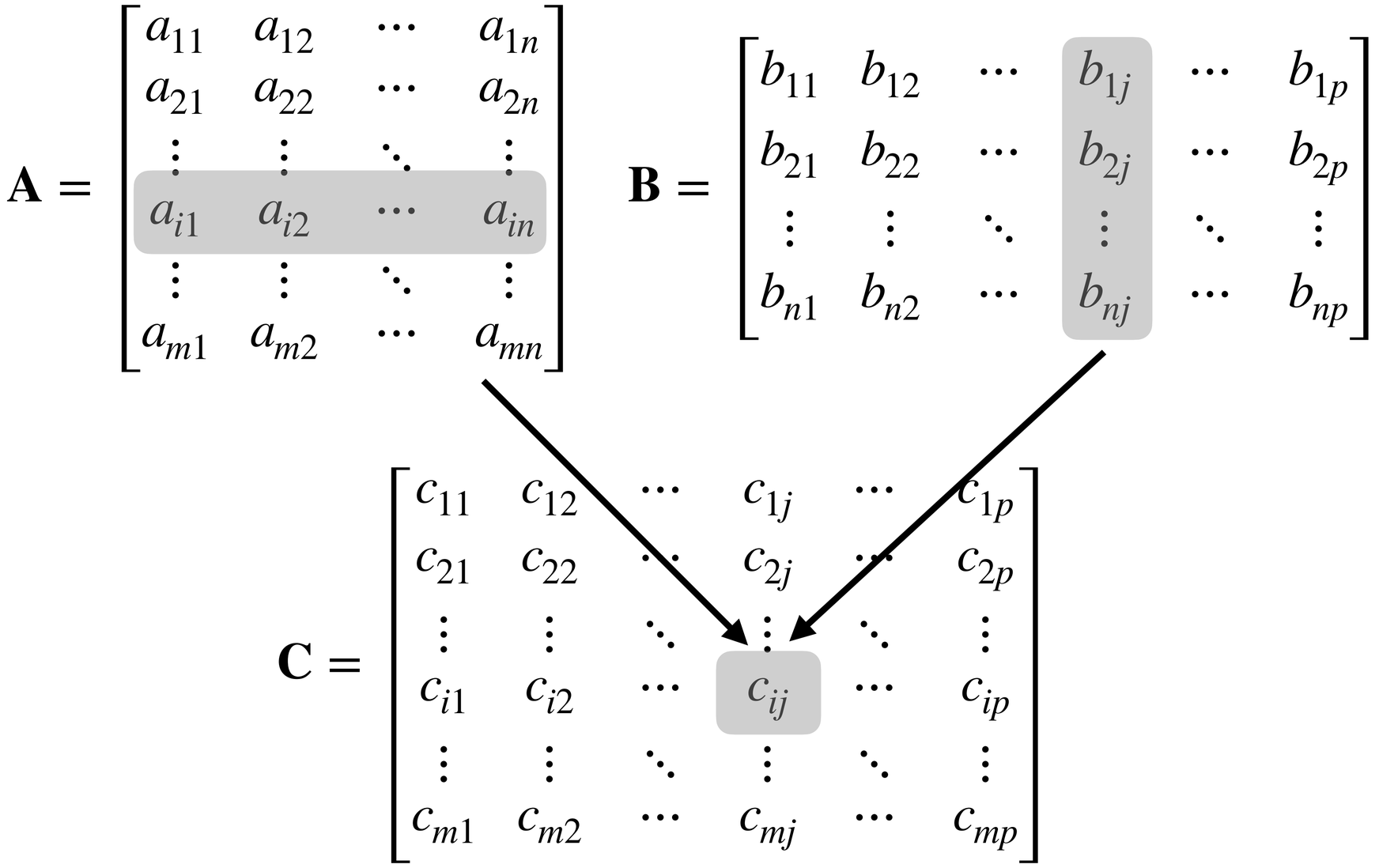
Intuitions

$$\mathbf{A} \in \mathbb{R}^{m \times n}$$

$$\mathbf{B} \in \mathbb{R}^{n \times p}$$

$$\mathbf{AB} \in \mathbb{R}^{m \times p}$$

$$\mathbf{AB} = C$$



Matrices

Identity matrix and inverse matrix

$$\mathbf{I_n} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

- Square matrix
- All elements on the diagonal from top-left to bottom-right are 1s
- everything else is 0
- $\mathbf{AI = IA = A}$

Matrices

Inverse and transpose

The inverse matrix $\mathbf{A^{-1}}$ of a square matrix $\mathbf{A}$ is defined such that $\mathbf{AA^{-1}} = \mathbf{A^{-1}A} = \mathbf{I}$, where $\mathbf{I}$ is the identity matrix

The transpose of a matrix is obtained by flipping the matrix over its main diagonal (from top-left to bottom-right), which swaps rows with columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}, \quad A^\top = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m} & a_{2m} & \cdots & a_{nm} \end{bmatrix}$$

Matrices

Properties

The inverse of a matrix multiplication is the product of the inverses in the reverse order $\quad (\mathbf{AB})^{-1} = \mathbf{B^{-1}A^{-1}}$

The transpose of a matrix multiplication is the product of the transposes in the reverse order $\quad (\mathbf{AB})^{T} = \mathbf{B^{T}A^{T}}$

Dot product with transpose $\quad \mathbf{uv} = \mathbf{v}^{T}\mathbf{u}$

For diagonal matrices:

$$\mathbf{D} = \mathbf{D}^{T} \qquad \mathbf{D} = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} \quad \mathbf{D^{-1}} = \begin{bmatrix} \frac{1}{d_1} & 0 & \cdots & 0 \\ 0 & \frac{1}{d_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{d_n} \end{bmatrix}$$

Matrices

Orthogonal matrices

A square matrix is orthogonal if its transpose is also its inverse:
$\mathbf{QQ^T = Q^TQ = I}$, where $\mathbf{I}$ is the identity matrix

Orthogonal matrices preserve length and relative angles

$$\|\mathbf{Qx}\|_2^2 = (\mathbf{Qx})^T(\mathbf{Qx}) = \mathbf{x}^T(\mathbf{Q}^T\mathbf{Q})\mathbf{x} = \mathbf{x}^T\mathbf{x} = \|\mathbf{x}\|_2^2$$

$$\mathbf{u} \cdot \mathbf{v} = \|u\|\|v\|cos\theta \qquad (\mathbf{Qv})^T(\mathbf{Qu}) = \mathbf{v}^T(\mathbf{Q}^T\mathbf{Q})\mathbf{u} = \mathbf{v}^T\mathbf{u} = u \cdot v$$